National Qualifications 2017

# X713/77/02

# Chemistry
# Section 1 — Questions

MONDAY, 8 MAY

9:00 AM – 11:30 AM

Instructions for the completion of Section 1 are given on *Page 02* of your question and answer booklet X713/77/01.

Record your answers on the answer grid on *Page 03* of your question and answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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**SECTION 1 — 30 marks**

**Attempt ALL questions**

**1.** All noble gases are characterised by the completion of the outermost orbital.
This orbital is

A an s-orbital
B a p-orbital
C a d-orbital
D an s or p-orbital.

**2.** The electronic configuration of an atom of X, in its ground state, is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$.
X is an atom of

A calcium
B scandium
C titanium
D vanadium.

**3.** Which line in the table could represent the four quantum numbers of an outer electron in an $Mg^{2+}$ ion?

| | *n* | *l* | *m* | *s* |
|---|---|---|---|---|
| A | 2 | 1 | −2 | −½ |
| B | 2 | 0 | 0 | +½ |
| C | 2 | 1 | −1 | +½ |
| D | 3 | 0 | 0 | −½ |

**4.** The coordination number of an ionic lattice can be determined by using the following equation.

$$\text{radius ratio} = \frac{\text{radius of positive ion}}{\text{radius of negative ion}}$$

| *Radius ratio* | *Coordination number* |
|---|---|
| less than 0·23 | 3 |
| 0·23–0·42 | 4 |
| 0·42–0·73 | 6 |
| greater than 0·73 | 8 |

What is the coordination number in zinc(II) sulfide?

A 3
B 4
C 6
D 8

**5.** What is the formula for the diaquatetrachlorocobaltate(II) ion?

A $[CoCl_4(H_2O)_2]^{2-}$
B $[CoCl_2(H_2O)_4]^{2-}$
C $[CoCl_4(H_2O)_2]^{2+}$
D $[CoCl_2(H_2O)_4]^{2+}$

**6.** Which of the following indicators is most suitable to use in a titration of dilute hydrochloric acid solution with dilute ammonia solution?

A Bromothymol blue
B Phenolphthalein
C Methyl orange
D Phenol red

**7.** The pH of a solution of benzoic acid with concentration 0·01 mol $l^{-1}$ is

A 1·1
B 2·0
C 3·1
D 5·2.

**8.** A reaction must be exothermic if

A both $\Delta G°$ and $\Delta S°$ are negative

B both $\Delta G°$ and $\Delta S°$ are positive

C $\Delta G°$ is negative

D $\Delta S°$ is positive.

**9.** For the reaction

$A + B \rightarrow C$

the following data were obtained.

| *Experiment* | *Initial concentration of A* (mol $l^{-1}$) | *Initial concentration of B* (mol $l^{-1}$) | *Initial rate of formation of C* (mol $l^{-1}$ $s^{-1}$) |
|---|---|---|---|
| 1 | 0·1 | 0·1 | 0·05 |
| 2 | 0·2 | 0·1 | 0·05 |
| 3 | 0·1 | 0·2 | X |

Given that the rate equation is

$\text{Rate} = k[B]^2$

the value of X will be

A 0·05

B 0·10

C 0·15

D 0·20.

**10.** The rate equation for the reaction between nitrogen monoxide and chlorine is

$\text{rate} = k[NO]^2[Cl_2]$

The units for the rate constant, k, in this reaction are

A $s^{-1}$

B mol $l^{-1}$ $s^{-1}$

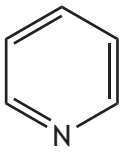
C $l\,mol^{-1}\,s^{-1}$

D $l^2\,mol^{-2}\,s^{-1}$.

**11.** Which of the following describes the bonding in ethane?

A $sp^2$ hybridisation with sigma bonds only.
B $sp^3$ hybridisation with sigma bonds only.
C $sp^2$ hybridisation with sigma and pi bonds.
D $sp^3$ hybridisation with sigma and pi bonds.

**12.** Pyridine has the following structure.




The number of sigma bonds in a molecule of pyridine is

A 3
B 6
C 11
D 12.

**13.** A racemic mixture is defined as

A a mixture of two enantiomers
B a pair of enantiomers mixed in equal proportions
C a mixture of two geometric isomers
D a pair of geometric isomers mixed in equal proportions.

**[Turn over**

**14.** $CH_3CH_2Br + NH_3 \rightarrow CH_3CH_2NH_2 + HBr$

$CH_3Br + OH^- \rightarrow CH_3OH + Br^-$

The nucleophiles in these two reactions are

A $CH_3Br$ and $NH_3$

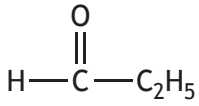
B $OH^-$ and $CH_3CH_2Br$

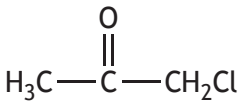
C $CH_3CH_2Br$ and $CH_3Br$

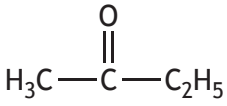
D $NH_3$ and $OH^-$.

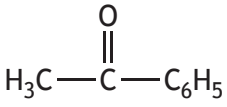
**15.** A compound X has a $GFM$ of less than 100 g.
Complete combustion of compound X produces carbon dioxide and water only.
Reduction of compound X produces a secondary alcohol.

Compound X is most likely to be

A 

B 

C 

D 

**16.**

| *Amine* | *Boiling point* (°C) |
|---|---|
| $C_2H_5N(CH_3)_2$ | 37·5 |
| $(C_2H_5)_2NH$ | 56·3 |
| $C_4H_9NH_2$ | 77·8 |

Based on the information in the table,

A the tertiary amine has the highest boiling point

B the secondary amine has the lowest boiling point

C the primary amine has a lower boiling point than the tertiary amine

D the secondary amine has a lower boiling point than the primary amine.

**17.** Compound Y reacts with the product of its own oxidation to form an ester.

Compound Y could be

A propanal

B propan-1-ol

C propan-2-ol

D propanoic acid.

**18.** Which of the following statements about benzene is correct?

A The benzene molecule is planar.

B Benzene does not react with electrophiles.

C Benzene readily undergoes nucleophilic attack.

D The benzene molecule contains carbon to carbon bonds of two different lengths.

**19.** Chlorine has two isotopes, $^{35}Cl$ and $^{37}Cl$.

These isotopes are present in a sample of 1,1,1-trichloroethane, $C_2H_3Cl_3$. The number of molecular ion peaks expected in the mass spectrum of 1,1,1-trichloroethane is

A 6

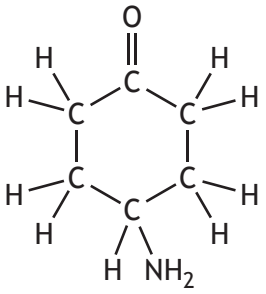
B 4

C 3

D 2.

**[Turn over**

**20.** The following substance was analysed using an infrared spectrometer.



The spectrum produced would **not** have a significant peak in the wave number range

A 1700–1680 $cm^{-1}$

B 2962–2853 $cm^{-1}$

C 3100–3000 $cm^{-1}$

D 3500–3300 $cm^{-1}$.

**21.** Antisense drugs are a group of medicines that act by binding to DNA to block the synthesis of some proteins.

Which line in the table is correct for antisense drugs?

| | *Classification* | *Receptor* |
|---|---|---|
| A | antagonist | DNA |
| B | antagonist | protein |
| C | agonist | DNA |
| D | agonist | protein |

**22.** Which of the following would be most suitable as a reagent in the gravimetric analysis of silver ions?

A Sodium nitrate

B Potassium sulfate

C Barium carbonate

D Ammonium chloride

**23.** Using colorimetry, the most appropriate filter for determining the concentration of green nickel ions, $Ni^{2+}(aq)$, in a solution, would be

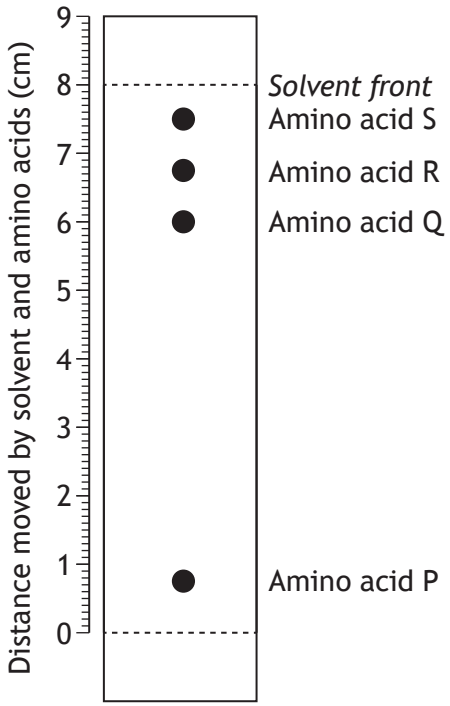
A 390 nm

B 490 nm

C 540 nm

D 680 nm.

**24.** The diagram shows a thin layer chromatogram for a mixture of amino acids.




Which amino acid has an $R_f$ value of approximately 0·75?

A Amino acid S

B Amino acid R

C Amino acid Q

D Amino acid P

**[Turn over**

**25.** Which line in the table shows the properties of the most suitable solvent to extract caffeine from an aqueous solution of tea?

| | | |
|---|---|---|
| A | Caffeine is more soluble in the solvent than it is in the tea solution. | The solvent is immiscible in the tea solution. |
| B | Caffeine is more soluble in the solvent than it is in the tea solution. | The solvent is miscible in the tea solution. |
| C | Caffeine is less soluble in the solvent than it is in the tea solution. | The solvent is miscible in the tea solution. |
| D | Caffeine is less soluble in the solvent than it is in the tea solution. | The solvent is immiscible in the tea solution. |

**26.** A series of titrations was performed to determine the concentration of vitamin C in a brand of fruit juice. A standard solution of the fruit juice was prepared and titrated with iodine solution.

Which of the following would be a suitable control experiment for this analysis?

A Titrate more samples from the same carton of fruit juice.

B Titrate a solution of pure vitamin C of known concentration.

C Titrate more samples from the standard solution of fruit juice.

D Titrate a sample from a different carton of the same brand of fruit juice.

**27.** $Ba(OH)_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaOH(aq)$

$50\,cm^3$ of $0{\cdot}010\,mol\,l^{-1}$ barium hydroxide solution were added to $50\,cm^3$ of $0{\cdot}010\,mol\,l^{-1}$ sodium sulfate solution.

The concentration of sodium hydroxide, in $mol\,l^{-1}$, in the resulting solution is

A 0·0010

B 0·010

C 0·020

D 0·10.

**28.** $1{\cdot}06 \times 10^{-2}$ moles of phenylamine, $C_6H_5NH_2$, react with 5·16 g of bromine.

Which equation shows the correct stoichiometry for this reaction?

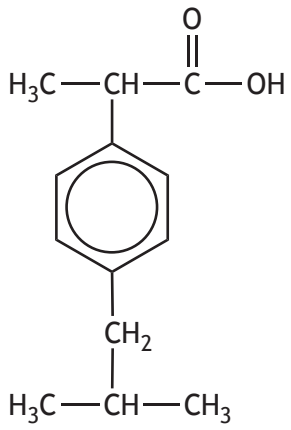
A $C_6H_5NH_2 + Br_2 \rightarrow C_6H_4BrNH_2 + HBr$

B $C_6H_5NH_2 + 2Br_2 \rightarrow C_6H_3Br_2NH_2 + 2HBr$

C $C_6H_5NH_2 + 3Br_2 \rightarrow C_6H_2Br_3NH_2 + 3HBr$

D $C_6H_5NH_2 + 4Br_2 \rightarrow C_6HBr_4NH_2 + 4HBr$

**29.** Ibuprofen is used for the relief of pain, fever and inflammation. A structural formula for ibuprofen is shown below.




If one tablet contains 300 mg of ibuprofen, approximately how many tablets can be manufactured from 1 mole of ibuprofen?

A $6{\cdot}73 \times 10^{2}$

B $6{\cdot}87 \times 10^{2}$

C $6{\cdot}73 \times 10^{-1}$

D $6{\cdot}87 \times 10^{-1}$

**[Turn over for next question**

**30.** The term accuracy is used to describe how close an experimental result is to the theoretical value. The term precision is used to describe how close a set of duplicate results are to each other.

Four students determined the percentage by mass of chlorine in $BaCl_2.2H_2O$.

Which of the following sets of results is both accurate and precise?

A 29·0%, 29·0%, 29·1%

B 29·1%, 28·2%, 29·9%

C 34·0%, 34·1%, 34·0%

D 34·0%, 34·3%, 33·8%

**[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET]**

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| | | | | | |
|---|---|---|---|---|---|
| | | | | | |

National Qualifications 2017

Mark

# X713/77/01

# Chemistry
# Section 1 — Answer Grid and Section 2

MONDAY, 8 MAY

9:00 AM – 11:30 AM

*X7137701*

**Fill in these boxes and read what is printed below.**

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

**Total marks — 100**

**SECTION 1 — 30 marks**

Attempt ALL questions.

Instructions for the completion of Section 1 are given on *Page 02*.

**SECTION 2 — 70 marks**

Attempt ALL questions.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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## SECTION 1 — 30 marks

The questions for Section 1 are contained in the question paper X713/77/02.

Read these and record your answers on the answer grid on *Page 03* opposite.

Use **blue** or **black** ink. Do NOT use gel pens or pencil.

1. The answer to each question is **either** A, B, C or D. Decide what your answer is, then fill in the appropriate bubble (see sample question below).

2. There is **only one correct** answer to each question.

3. Any rough working should be done on the additional space for answers and rough work at the end of this booklet.

**Sample Question**

To show that the ink in a ball-pen consists of a mixture of dyes, the method of separation would be:

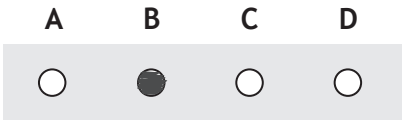
A fractional distillation

B chromatography

C fractional crystallisation

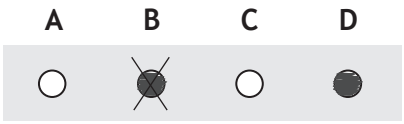
D filtration.

The correct answer is **B** — chromatography. The answer **B** bubble has been clearly filled in (see below).

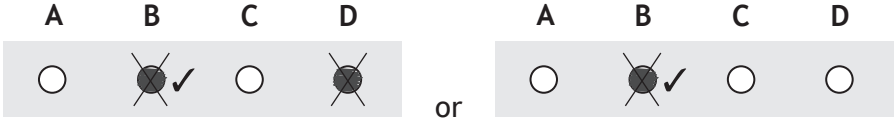



**Changing an answer**

If you decide to change your answer, cancel your first answer by putting a cross through it (see below) and fill in the answer you want. The answer below has been changed to **D**.




If you then decide to change back to an answer you have already scored out, put a tick (✓) to the **right** of the answer you want, as shown below:




or

* X 7 1 3 7 7 0 1 0 2 *

## SECTION 1 — Answer Grid

*0BJ30AD1*

| | A | B | C | D | | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| 1 | ○ | ○ | ○ | ○ | 16 | ○ | ○ | ○ | ○ |
| 2 | ○ | ○ | ○ | ○ | 17 | ○ | ○ | ○ | ○ |
| 3 | ○ | ○ | ○ | ○ | 18 | ○ | ○ | ○ | ○ |
| 4 | ○ | ○ | ○ | ○ | 19 | ○ | ○ | ○ | ○ |
| 5 | ○ | ○ | ○ | ○ | 20 | ○ | ○ | ○ | ○ |
| 6 | ○ | ○ | ○ | ○ | 21 | ○ | ○ | ○ | ○ |
| 7 | ○ | ○ | ○ | ○ | 22 | ○ | ○ | ○ | ○ |
| 8 | ○ | ○ | ○ | ○ | 23 | ○ | ○ | ○ | ○ |
| 9 | ○ | ○ | ○ | ○ | 24 | ○ | ○ | ○ | ○ |
| 10 | ○ | ○ | ○ | ○ | 25 | ○ | ○ | ○ | ○ |
| 11 | ○ | ○ | ○ | ○ | 26 | ○ | ○ | ○ | ○ |
| 12 | ○ | ○ | ○ | ○ | 27 | ○ | ○ | ○ | ○ |
| 13 | ○ | ○ | ○ | ○ | 28 | ○ | ○ | ○ | ○ |
| 14 | ○ | ○ | ○ | ○ | 29 | ○ | ○ | ○ | ○ |
| 15 | ○ | ○ | ○ | ○ | 30 | ○ | ○ | ○ | ○ |

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**SECTION 2 — 70 marks**

**Attempt ALL questions**

**1.** Many of the paints used by artists contain cadmium compounds.

The presence of cadmium in a paint sample can be detected by atomic emission spectroscopy.

(a) (i) Explain how a line is produced in an emission spectrum. **2**

(ii) Explain why there is a series of lines at discrete wavelengths in the emission spectrum of cadmium. **1**

(b) The cadmium emission spectrum has a line at 644 nm.

Calculate the energy, in kJ $mol^{-1}$, associated with this wavelength. **2**

**2.** Phosphorus forms different compounds with chlorine.

(a) When heated, phosphorus pentachloride dissociates to form phosphorus trichloride and chlorine.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) \qquad \Delta H = +124\ kJ\ mol^{-1}$$

In an experiment to determine the equilibrium constant, $K$, 0·100 mol of $PCl_5$ was placed in a sealed 1·00 litre flask and heated to 250 °C. At equilibrium 0·0420 mol of $PCl_3$ had been formed.

(i) Calculate the equilibrium constant, $K$, for the reaction at 250 °C. **3**

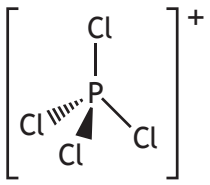
(ii) The temperature of the equilibrium mixture was increased to 400 °C.

Explain the effect of this change in temperature on the value of the equilibrium constant, $K$. **2**

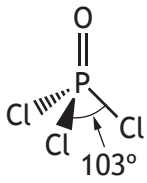
**2. (continued)**

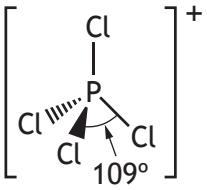
(b) In the solid state, phosphorus pentachloride is ionic and has the formula $[PCl_4]^+[PCl_6]^-$.

(i) The three-dimensional structure for the $[PCl_4]^+$ ion is shown. Complete the table for the $[PCl_6]^-$ ion. **1**

| *Phosphorus species* | *Three-dimensional structure* |
|---|---|
| $[PCl_4]^+$ |  |
| $[PCl_6]^-$ | |

(ii) Phosphorus oxychloride, $POCl_3$, has a similar three-dimensional structure to the $[PCl_4]^+$ ion as shown.




Suggest a reason why the bond angle in the $POCl_3$ molecule is less than the bond angle in the $[PCl_4]^+$ ion. **1**

**3.** Zinc is often found in nature together with lead in sulfide ores. Different industrial processes can be used for the production of zinc metal. One of these is an electrolytic process and another is a thermal process.

(a) In the electrolytic process, zinc sulfide is converted into zinc oxide by roasting in a furnace at 1300 K.

$$2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$$

The data in the table refers to this reaction.

| *Substance* | $\Delta H_f°$ (kJ mol$^{-1}$) | $S°$ (J K$^{-1}$ mol$^{-1}$) |
|---|---|---|
| ZnS(s) | −206 | 58 |
| $O_2$(g) | 0 | 205 |
| ZnO(s) | −350 | 44 |
| $SO_2$(g) | −297 | 248 |

(i) For the conversion of zinc sulfide into zinc oxide, use the data in the table to calculate:

(A) $\Delta H°$, in kJ mol$^{-1}$; **1**

(B) $\Delta S°$, in J K$^{-1}$ mol$^{-1}$. **1**

(ii) Calculate the theoretical temperature, in K, above which the reaction is no longer feasible. **2**

**3. (continued)**

(b) In the thermal process, a mixture of zinc oxide and lead oxide is reacted with carbon in a furnace at a temperature of 1200 K.

Data for the metals and metal oxides are shown in the table below.

| *Substance* | *Density* ($g\ cm^{-3}$) | *Melting point* (K) | *Boiling point* (K) |
|---|---|---|---|
| Zn | 7·1 | 693 | 1181 |
| Pb | 11·3 | 600 | 2024 |
| ZnO | 5·6 | 2248 | 2633 |
| PbO | 9·5 | 1161 | 1808 |

By considering all the information, suggest how a sample of zinc metal and a sample of lead metal could each be removed from the furnace. **2**

**[Turn over**

**4.** Transition metals, such as vanadium and copper, can have variable oxidation states and a wide range of uses.

(a) Vanadium dioxide, $VO_2$, can be used to coat glass.

(i) State the oxidation number of vanadium in $VO_2$. **1**

(ii) Using orbital box notation, write the electronic configuration, in terms of s, p and d orbitals, for the vanadium ion in $VO_2$. **1**

(b) Vanadium(III) ions can react with iron(III) ions in solution.

$$V^{3+}(aq) + Fe^{3+}(aq) \rightarrow V^{4+}(aq) + Fe^{2+}(aq)$$

(i) The reaction is first order with respect to both $V^{3+}(aq)$ and $Fe^{3+}(aq)$.
Write the rate equation for this reaction. **1**

**4. (b) (continued)**

(ii) In the presence of a $Cu^{2+}(aq)$ catalyst, the reaction mechanism is:

$$Cu^{2+}(aq) + V^{3+}(aq) \xrightarrow{\text{slow}} Cu^{+}(aq) + V^{4+}(aq)$$

$$Cu^{+}(aq) + Fe^{3+}(aq) \xrightarrow{\text{fast}} Cu^{2+}(aq) + Fe^{2+}(aq)$$

(A) State the order of the reaction with respect to $Fe^{3+}(aq)$ when a $Cu^{2+}(aq)$ catalyst is present.

Explain your answer. **2**

(B) Explain why $Cu^{2+}(aq)$ can be described as a homogeneous catalyst in this reaction. **1**

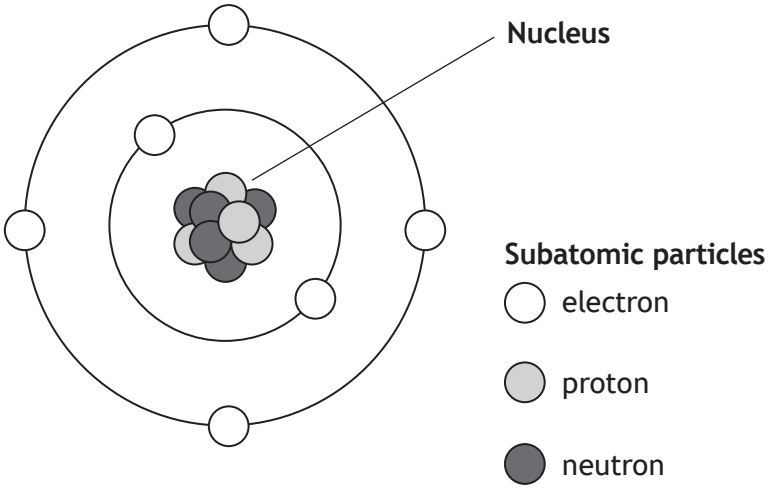
(c) In the Middle Ages, Damascus steel was used for making sword blades.

The steel from a sword blade of mass 1300 g was found to have a vanadium concentration of 71 ppm.

Calculate the total mass of vanadium present in the sword blade. **2**

**[Turn over**

**5.** A simple model of an atom is shown.




This simplistic model can be useful to help explain bonding but it is also misleading, as the structure of the atom and bonding are more complicated.

**Using your knowledge of chemistry**, discuss the strengths and weaknesses of this simple model compared to the concepts of atomic structure and bonding at Advanced Higher level. **3**

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**5. (continued)**

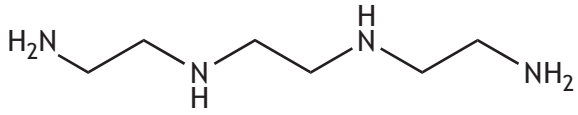
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**6.** Wilson's disease is a rare genetic disorder which results in a build-up of copper ions in the body. Unmetabolised copper ions are toxic, leading to health complications.

Copper ions can be removed by reaction with trientine.



trientine

(a) Trientine is a tetradentate ligand that reacts with copper(II) ions in a 1:1 ratio to form a complex ion which can then be removed from the body.

(i) Ligands form dative covalent bonds with metal ions.

State what is meant by a dative covalent bond. **1**

(ii) Draw a structural formula for the complex ion. **1**

**6. (continued)**

(b) Zinc ethanoate can also be used to treat Wilson's disease.

(i) Zinc ethanoate can be prepared from zinc hydroxide and ethanoic acid.

Name this type of reaction. **1**

(ii) Zinc ethanoate is a salt of a weak acid.

State what is meant by a weak acid. **1**

(iii) A student carried out an experiment to determine the value of $y$ in hydrated zinc ethanoate, $Zn(CH_3COO)_2.yH_2O$. A 5·00 g sample was heated until all the water was removed and a constant mass of 4·18 g was obtained.

(A) Name the piece of apparatus that should be used to store the zinc ethanoate while cooling. **1**

(B) Calculate the value of $y$. **2**

(C) The student repeated the experiment with a second sample of hydrated zinc ethanoate.

The student's calculations were correct but the value of $y$ was found to be different from the expected value.

Suggest a reason for this difference. **1**

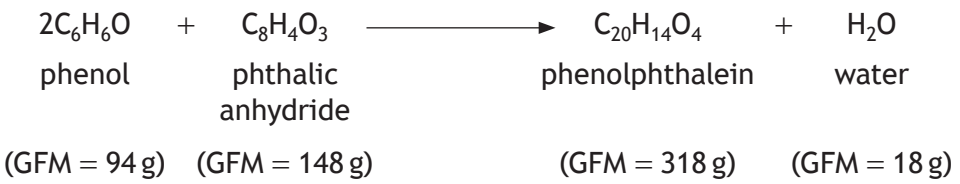
**7.** Phenolphthalein is an indicator that can be prepared by the reaction of phenol with phthalic anhydride.

(a) Phenolphthalein prepared by this method can have insoluble impurities present. It can be purified by recrystallisation from an aqueous ethanol solution.

(i) Outline the steps that should be carried out to recrystallise phenolphthalein. **2**

(ii) Name a technique that could be used to determine if the recrystallised phenolphthalein is pure. **1**

(b) The equation for the reaction is given below.




0·96 g of phenol is reacted with 1·05 g phthalic anhydride.

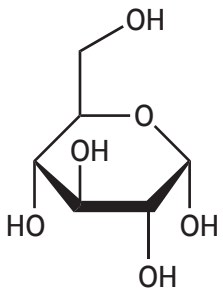
In an experimental procedure, the percentage yield of phenolphthalein was 58%.

Calculate the mass, in grams, of phenolphthalein produced. **3**

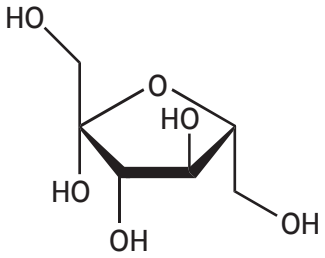
**8.** High levels of sugar in foods are associated with obesity. Sugars are also the basis of many medicines.

(a) Glucose-fructose syrup is a type of sugar that is added to many foods. It is a mixture of glucose and fructose.

The ring structures of glucose and fructose are shown below.




glucose




fructose

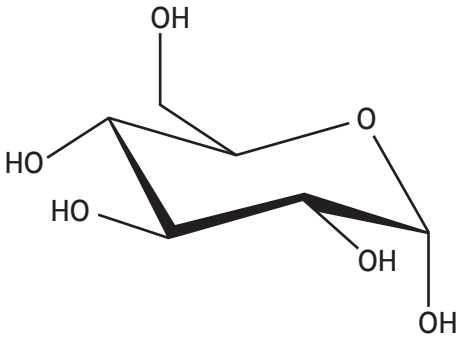
(i) Write the molecular formula for fructose. **1**

(ii) Suggest, with reference to the structures, how $^{1}H$ NMR spectroscopy could be used to distinguish between glucose and fructose. **1**

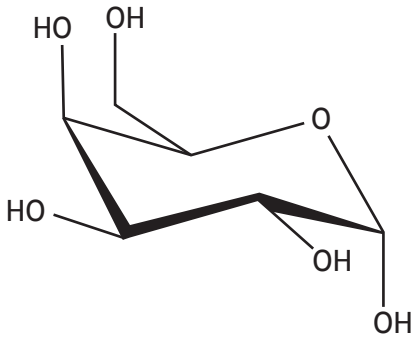
**[Turn over**

**8. (continued)**

(b) A more accurate representation of the structure of glucose, and its geometric isomer galactose, is shown below.



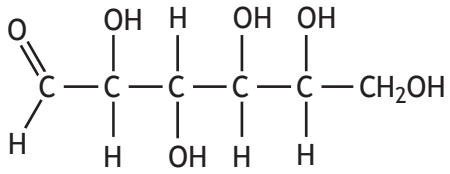

glucose




galactose

With reference to the structures shown, explain why sugars such as glucose and galactose have geometric isomers. **1**

(c) The ring structure of glucose exists in equilibrium with its open-chain structure. The diagram below shows the open-chain structure of one optical isomer of glucose called D-glucose.




D-glucose

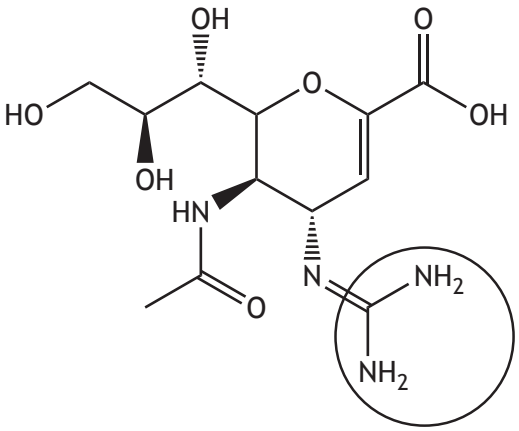
(i) State the number of chiral centres in D-glucose. **1**

(ii) Draw an open-chain structural formula for an optical isomer of D-glucose. **1**

**8. (continued)**

(d) Relenza is a sugar-based medicine used to treat the flu virus. It acts by attaching to an enzyme active site on the virus.

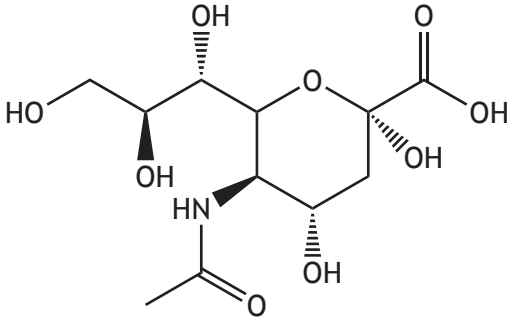
The structure of Relenza is shown.




Relenza

(i) Suggest how the functional groups circled on the Relenza molecule would bind with part of the enzyme active site. **1**

(ii) The structure of the natural active compound, sialic acid, is shown.



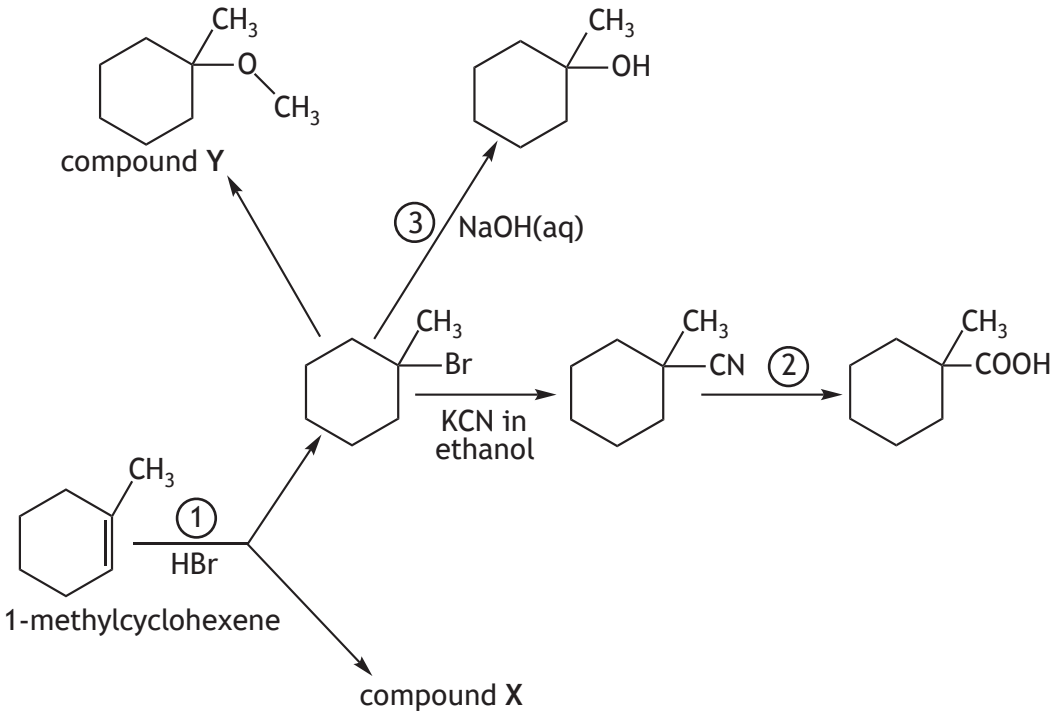

sialic acid

Sialic acid binds to the same part of the enzyme active site as Relenza.

Circle the functional groups on the sialic acid molecule which are most likely to bind with the enzyme active site.

(An additional diagram, if required, can be found on *Page 28*) **1**

**9.** A student devised the following reaction scheme starting with 1-methylcyclohexene.




(a) In reaction ①, 1-methylcyclohexene reacts with HBr to produce two compounds.

(i) Draw a structural formula for compound **X**. **1**

(ii) Reaction ① obeys Markovnikov's rule.

Explain, with reference to the carbocation intermediate, why compound **X** is the minor product in this reaction. **1**

**9. (continued)**

(b) Suggest a reagent that could be used in reaction ②. **1**

(c) Reaction ③ is likely to undergo an $S_N1$ mechanism.
Using curly arrow notation, draw the mechanism for this reaction. **2**

(d) Name compound **Y**. **1**

**[Turn over**

**10.** An active ingredient in many stain removing products is the oxidising agent hydrogen peroxide, $H_2O_2$.

(a) In an experiment to determine the concentration of hydrogen peroxide present in a stain remover a student carried out a titration with acidified permanganate solution.

$$2MnO_4^- + 5H_2O_2 + 6H^+ \rightarrow 2Mn^{2+} + 5O_2 + 8H_2O$$

5·0 $cm^3$ of stain remover was pipetted into a 100 $cm^3$ standard flask and made up to the mark with distilled water.

20·0 $cm^3$ samples were titrated with 0·030 mol $l^{-1}$ permanganate solution until a permanent pink colour remained. The results are shown in the table.

| | 1st titration | 2nd titration | 3rd titration |
|---|---|---|---|
| Initial burette reading ($cm^3$) | 0·3 | 19·2 | 0·2 |
| Final burette reading ($cm^3$) | 19·2 | 37·7 | 18·8 |
| Volume used ($cm^3$) | 18·9 | 18·5 | 18·6 |

(i) Calculate the number of moles of hydrogen peroxide in 20·0 $cm^3$ of the diluted solution of stain remover. **2**

(ii) Calculate the concentration, in mol $l^{-1}$, of hydrogen peroxide in the undiluted stain remover. **2**

**10. (a) (continued)**

(iii) The concentration of hydrogen peroxide determined by the student was less than the concentration stated on the label for the stain remover.

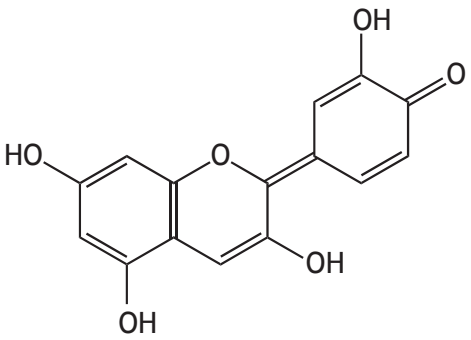
One possible source of error could be an inaccurate concentration of the permanganate solution.

Describe how the student would confirm the concentration of the permanganate solution. **1**

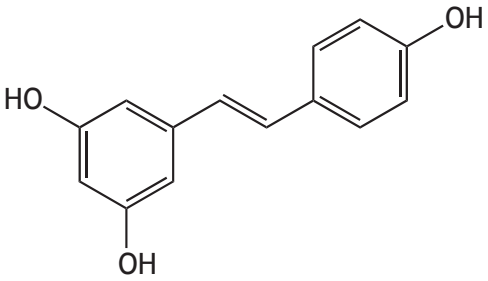
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**10. (continued)**

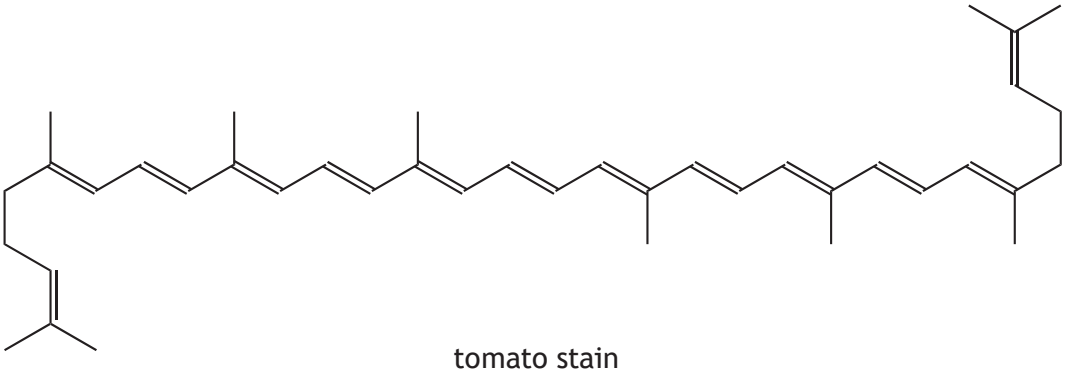
(b) Some of the molecules thought to be responsible for the colour of stains are shown.




blackcurrant stain




red wine stain



tomato stain

**Using your knowledge of chemistry**, suggest how the chemicals in a stain remover might work on these stains. **3**

**10. (b) (continued)**

**[Turn over**

**11.** Nutmeg is a seed that is commonly used as a spice in cooking. The flavour of nutmeg is due to a number of different compounds.

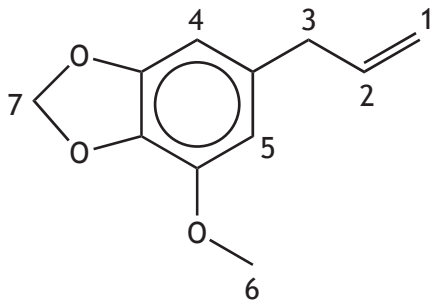
(a) The oil in nutmeg, trimyristin, can be easily extracted and purified.

In an experiment to extract trimyristin, a student refluxed nutmeg powder in a suitable solvent. Removal of the solvent produced an impure sample of solid trimyristin.

(i) Suggest why an ether could be a suitable solvent to extract trimyristin oil from nutmeg. **1**

(ii) Suggest why the mixture was heated under reflux. **1**

(b) Myristicin is another compound that can be isolated from nutmeg. $^{1}H$ NMR analysis showed there to be seven proton environments and these are numbered on the skeletal formula shown.



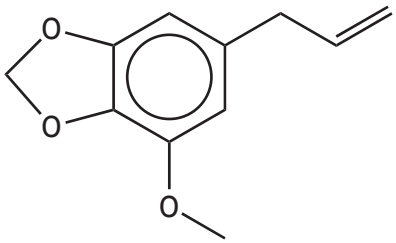

myristicin

(i) Suggest a possible chemical shift for the peak arising due to proton environment 1. **1**

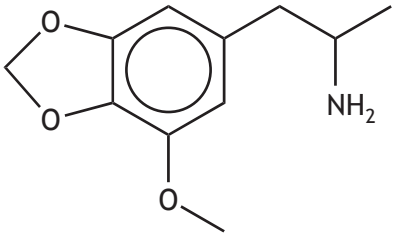
(ii) Identify a proton environment which would produce a doublet in the $^{1}H$ NMR spectrum. **1**

**11. (continued)**

(c) Myristicin can be converted into compound X in **two** steps.



myristicin




compound X

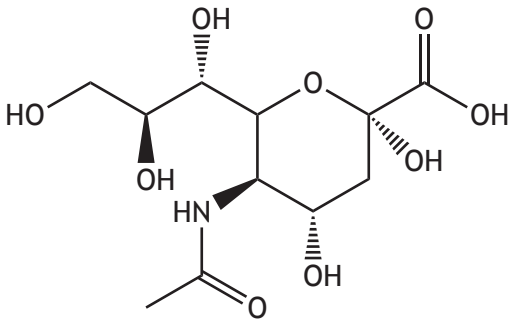
Suggest the type of reaction occurring at **each** step. **2**

Step 1:

Step 2:

**[END OF QUESTION PAPER]**

**ADDITIONAL DIAGRAM FOR USE IN QUESTION 8 (d) (ii)**




sialic acid

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